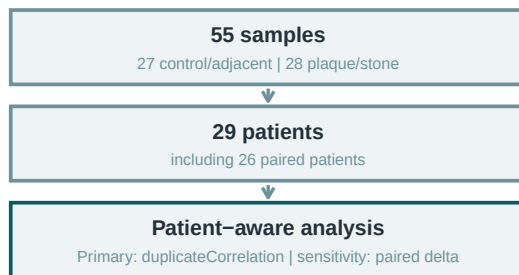
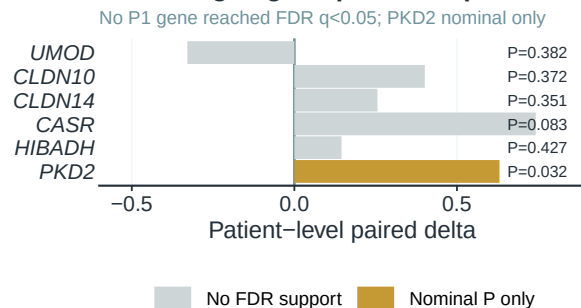


A. Patient-aware GSE73680 disease-context design

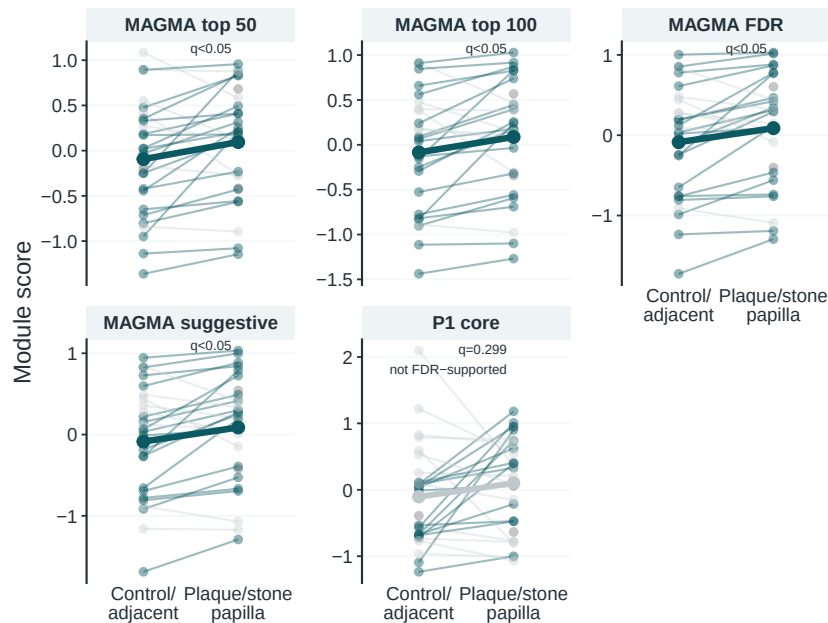


B. P1 single-gene paired response



C. MAGMA modules show paired plaque/stone papilla shifts

Each line represents one paired patient



D. Random-set benchmark and robustness

Teal: >95th percentile

